Name……...Centre/Index number U……../……..

UACE

P 530/3

JUNE, 2019

$\mathbf{3\frac{1}{4}}$ ***HOURS***

Uganda Advanced Certificate of Education

INTERNAL MOCK EXAMINATIONS

BIOLOGY PRACTICAL

Paper 3

TIME ALLOWED: 3HOURS & 15 MINUTES

INSTRUCTIONS:

- *This paper consists of three questions.*
- *Answer **ALL** questions.*
- *Write your answers in the spaces provided. Additional sheets of paper must not be inserted in this booklet.*
- *You are not allowed to start working within the first 15 minutes. You are advised to use this time to read through the paper and ensure that you have all the apparatus, chemicals and specimens you may require.*

For Examiner's use ONLY

Question	**Mark**	**Examiner's initial**
1		
2		
3		
TOTAL		

1. You are provided with specimen **K** which is freshly killed. Examine the labial palps and maxillary palps of the specimen using hand lens.
   a) State the observable differences between the labial palps and maxillary palps of the specimen. (04marks)

Labial palp	**Maxillary palp.**

   b) Carefully cut off the head from its base and drop it in a glass beaker containing $30cm^3$ of boiling sodium hydroxide solution and a pin for 2 minutes. After, transfer the head into another beaker containing cold water and wash off the sodium hydroxide. Carefully detach the maxilla from the rest of the structures of the mouth. Place the maxilla on a glass slide and observe under medium power of microscope.
      i) Draw and label the structures of the maxillae in the space below.(10 marks)

ii) State two adaptations of the maxilla to its function (02marks)

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c) Dissect the specimen to expose the internal structures. Displace the alimentary canal to the right and remove any fatty materials obscuring the structures on the ventral cuticle. Draw and label the

i) Structures of the alimentary canal that constitute the hind gut, midgut, and back of the foregut

ii) Structures on the ventral cuticle anterior to the last segment. (18 marks)

d) Cut out the gizzard and the crop from the alimentary canal. Using a pair of scissors, cut the crop and gizzard longitudinally, wash off any materials from their walls. Examine the inner surface of both the gizzard and crop.

i) Describe how the structures of the inner surface of the gizzard and crop suit them to their functions (03marks)

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ii) Draw and label the longitudinal section of the structure described above (3marks)

2. You are provided with specimens **P_1** and **P_2**. Specimen **P_1** and **P_2** were obtained from the same plant but have been treated under different conditions for some days.

Solutions **W** and **N** are common laboratory reagents. Carry out tests on **P_1** and **P_2** and investigate the effect of **W** and **N** on **P_1** and **P_2**.

Peel specimen **P_1** and **P_2** using a knife, cut out a cube measuring 3cmX3cmX3cm from specimen **P_1**. Crush the cut piece in a motor using a pestle. Add 6cm^3 of distilled water to the fine paste and stir well. Allow to settle and decant to obtain the extract **P1**.

Repeat the above procedure using specimen **P_2** to obtain extract **P_2**.

a) Carryout tests on the extracts **P_1** and **P_2** as indicated in the table.

Table 1 **(15 marks)**

Test	Observation	Conclusion
Starch test		
Reducing sugar test		
Biuret test		

b) Label four test tubes 1, 2, 3, and 4. Treat the test tubes as shown below.

To test tube1 and test tube 2, add 1cm^3 of extract **P_1**

To test tube3 and test tube 4, add 1cm^3 of extract **P_2**.

To test tube1and test tube 3, add 1cm^3 of solution **W** while to test tube2 and test tube 4 add 1cm^3 of solution **N**. Write your observations and deductions in the table below.

Table 2 **(8marks)**

Test tube	Observation	Deduction
Test tube1		
Test tube2		
Test tube3		
Test tube4		

c) Obtain a thin slice from specimen **P_1** using a razor blade and another thin slice from one of the bud on specimen **P_2**. Mount each section on a microscopic slide and irrigate with a drop of iodine in each case. Wait for 3minutes and thereafter observe under medium power of a microscope and state the differences seen between the sections in the table below.(04marks)

Table 3

Section from specimen P1	**Section from specimen P2**

d) Explain the results in table (06marks)

i) Table 1

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ii) Table 2

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iii) Table 3

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3. You are provided with specimens **R, S, T, U** and **V**. Cut specimen **R** longitudinally and the rest of the specimens should be cut transversally.

   a) Describe the seed arrangement in each of the specimens cut.(10 marks)

      i) **R**

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      ii) **S**

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      iii) **T**

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      iv) **U**

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      v) **V**

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   b) State two observable internal differences and two similarities between specimen **T** and **U**.

      Differences (1 mark)

Specimen T	**Specimen U**

Similarities (1 mark)

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c) Draw and label a transverse section of specimen **T** (10 marks)

d) Using only the internal features of the specimens construct a dichotomous key to identify them. (05marks)

END

Confidential

Each candidate should be provided with the following

Specimen R- bean pod

S-an orange

T-cucumber

U-green paper

V-avocado

Freshly killed cockroach labelled K

Specimen P1-fresh irish potato

Specimen P2- sprouting irish potato

Solution W-5%hydrogen peroxide solution

Solution N- 1%hydrogen peroxide solution

Knife

A microscope

Microscope slide and cover slip

Distilled water

10ml measuring cylinder

Heat source

Reagent for food tests

10%sodium hydroxide solution

A small pin

Test tubes

A motor and pestle

Labels